

y.K class - 2

Find the power series of the DE, $y'' + xy' + (x+2)y = 0$ in power of x .

Soln :

Given DE, $y'' + xy' + (x+2)y = 0$ — (1)

Here, $x_0 = 0$ is an ordinary point of Eq (1)

Let the solution of Eq (1) is $y = \sum_{n=0}^{\infty} C_n (x - x_0)^n = \sum_{n=0}^{\infty} C_n x^n$ — (2)

Putting the values of y, y' in Eq (2) $\Rightarrow$

$y' = \sum_{n=1}^{\infty} n C_n x^{n-1}$ — (3), $y'' = \sum_{n=2}^{\infty} n(n-1) C_n x^{n-2}$ — (4)

$\sum_{n=2}^{\infty} n(n-1) C_n x^{n-2} + \sum_{n=0}^{\infty} n C_n x^n + \sum_{n=0}^{\infty} (n+2) C_n x^n = 0$

$\Rightarrow \sum_{n=2}^{\infty} (n+2)(n+1) C_{n+2} x^n + \sum_{n=2}^{\infty} n C_n x^n + \sum_{n=2}^{\infty} (n-2) C_n x^n + 2 \sum_{n=0}^{\infty} C_n x^n = 0$

Now, we take the common range from $n=2$ to ∞

$$2C_2 + 6C_3x + \sum_{n=2}^{\infty} (n+2)(n+1)C_{n+1}x^n + 2C_1x + 2C_0$$

$$2 \sum_{n=2}^{\infty} C_n x^n = 0$$

$$\Rightarrow (2C_2 + 2C_0) + (6C_3 + 3C_1)x + \sum_{n=2}^{\infty} [(n+2)(n+1)C_{n+2} + (n+2)C_n + C_{n-2}] x^n = 0$$

This leads to the conditions,

$$2C_2 + 2C_0 = 0 \quad - (5)$$

$$3C_1 + 6C_3 = 0 \quad - (6)$$

from (5) & (6) $C_2 = -C_0$

$$C_3 = -\frac{1}{2}C_2$$

and $[(n+2)(n+1)C_{n+2} + (n+2)C_n + C_{n-2}] = 0 ; n \geq 2$

$$C_{n+2} = \frac{(n+2)C_n + C_{n-2}}{(n+2)(n+1)} \quad - (7) \text{ which is called recurrence}$$

formula .

$$n=2$$

$$C_4 = - \frac{4C_2 + C_0}{12} = - \frac{-3C_0}{12} = \frac{1}{4} C_0 [C_2 = C_0]$$

$$n=3 \quad C_5 = \frac{3}{40} C_1 \dots$$

$$C_6 = \dots$$

$$C_7 = \dots \text{ etc}$$

Substituting this values of C_2, C_3, C_4 and $C_5 \dots$

in $E_2(2)$ we get.

$$y = \sum_{n=0}^{\infty} C_n x^n$$

$$y = C_0 + C_1 x - C_0 x^2 - \frac{1}{2} C_1 x^3 + \frac{1}{4} C_0 x^4 + \frac{3}{40} C_1 x^5$$

$$y = C_0 \left(1 - x^2 + \frac{1}{4} x^4 - \dots \right) + C_1 \left(x - \frac{1}{2} x^3 + \frac{3}{40} x^5 - \dots \right)$$

— (2)

which gives the solution of the DE (1) in power of x .

Problem - 1 :

Find a power series solution of the initial value problem $(x-1)y'' + 3xy' + 2xy = 0$.

Subject to the conditions $y(0) = 4$, $y'(0) = 6$.

Series Solution (Frobenius method)

~~Prob~~ Use the method of Frobenius to ~~the~~ find

the solution of the DE, $2x^2y'' - xy' + (x-5)y = 0$

~~Soln~~ Given DE $\Rightarrow 2x^2y'' - xy' + (x-5)y = 0$ — (1)

$$y'' - \frac{1}{2x}y' + \frac{x-5}{2x^2}y = 0$$

Here $x=0$ is a regular singular point.

Let the solution of $E_2(1)$,

$$y = \sum_{n=0}^{\infty} C_n (x-0)^{n+r} ; C_0 \neq 0 \quad \text{--- (2)}$$